



Methodological PLS-PM Framework for SDGs System

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Accepted: 8 January 2020 / Published online: 20 January 2020
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Abstract

Sustainability is the biggest challenge of our generation, because civilization has reached a point where natural resources are in rapid decline. It's a complex multidimensional phenomenon, which was studied for couple decades already. Nowadays different social concepts, such as sustainability, but also quality of life, satisfaction, are difficult and complex to define. The main problem for researchers is to find appropriate tools to obtain a composite indicator able to synthesize and represent these phenomena. The work focuses on building a system of composite indicators of Sustainability through to Structural Equation Modeling, specifically with the use of Partial Least Squares-Path Modeling. In recent years many advances have been developed, in the context of these models to solve some problems related to the role that the composite indicators play within that system; in particular on the aspects linked to the high level of abstraction, when a composite indicator is manifold, lacks its own manifest variables and is described by various underlying blocks. The aim of this paper is to demonstrate how these recent developments in Partial Least Squares-Path Modeling could help you to build a SDGs system and to provide a better measure of this complex social phenomenon.

Keywords Composite indicators · Higher-order construct · PLS-path modeling · Sustainable development goals

1 Introduction

The concept of sustainability and particularly of sustainable development figure among the most ambiguous and controversial in the literature. Sustainability has already been a popular term in the field of environmental economics. However, its definition is still ambiguous.

The term “sustainability” was first proposed in 1980 by the International Union for Conservation of Nature and Natural Resources (International Union for Conservation of Nature and World Wildlife Fund 1980). One of the important concluding remarks of this report was that *human beings, in their quest for economic development and enjoyment of the riches of nature, must come to terms with the reality of resource limitation and the carrying capacities of ecosystems, and must take account of the needs of future generations.* The

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Brundtland Report provides the most popular notion of sustainability (sustainable development):¹ development that meets the needs of the present without compromising the ability of future generations to meet their own needs (WCED, SPECIAL WORKING SESSION 1987). The definition provided by the Brundtland Report is a characteristic definition of sustainability (Bithas and Christofakis 2006).

Then there were several other milestones such as Rio Earth Summit in 1992. An important achievement of the summit was an agreement on the Climate Change Convention which in turn led to the Kyoto Protocol (1997) and the Paris Agreement (United Nations Development Programme 1992).

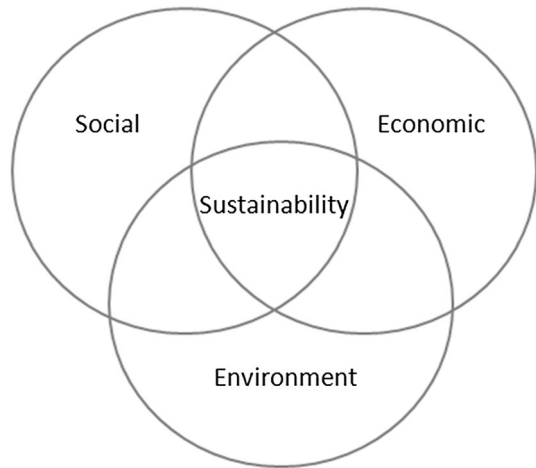
Measuring sustainable development and setting indicators for assessment and continuous improvement of the development of nations were discussed for the first time at that Rio Earth Summit, also known as Agenda 21, which called for development of “systems for monitoring and evaluation of progress towards achieving sustainable development by adopting indicators that measure changes across economic, social and environmental dimensions” (United Nations Development Programme 1992). Ever since, a wide range of indices have been proposed, each aiming to rank countries based on their achievements across different sets of indicators (Strezov et al. 2017).

Another important step in the history was the adoption of Millennium Development Goals (MDGs) (United Nations 2009). The MDGs mark a historic and effective method of global mobilisation to achieve a set of important social priorities worldwide. They express widespread public concern about poverty, hunger, disease, unmet schooling, gender inequality, and environmental degradation (Sachs 2012).

In 2012 Rio+20 Summit aimed at reconciling the economic and environmental goals of the global community (United Nations 2015b). It has issued a report recommending that the world adopt a set of Sustainable Development Goals (SDGs) which will build upon the MDGs and converge with the post 2015 development agenda. In fact, the conference on sustainable development, that led to United Nations 2030 Agenda, calls for an extensive set of global indicators, which consists of 17 Goals and 169 Targets as a result of joint work of all stakeholders: academia, civil society and government’s official delegates (United Nations 2015a). According to Sachs (2012), the SDGs are an important idea, and could help finally to move the world to a sustainable trajectory. The detailed content of the SDGs, if indeed they do emerge in upcoming diplomatic processes, is very much up for discussion and debate. In his paper, Sachs believes that the SDGs can benefit from both the successes and the shortfalls of the MDGs. The successes are notable. Unlike many UN goals, the MDGs are still very much with us almost 12 years after their adoption. Moreover the success of the SDGs will need societies worldwide to invest adequately in their success. Sustainable development is the only viable path for humanity, but it will not be achieved unless a small part of consumption spending is turned into investments for long-term survival. Sachs (2012). Based on literature review, we can see that it is difficult to define sustainability precisely and convincingly. Moreover, from this review of definitions, sustainability consists of two main elements: the triple bottom line as an abstract notion of environmental, social and economic processes and over time preservation or intergenerational equity (Mori and Christodoulou 2012). The concept of development and growth was studied by Solow (1994) and Daly et al. (1994), who argue that sustainability is linked not only to the economic component but also to the social and environmental dimensions.

¹ The concept of sustainable development is quite different from that of sustainability in that the word “development” clearly points to the idea of change, of directional and progressive change (Gallopín 2003).

Fig. 1 The graph of Sustainability



So, starting from these considerations, the sustainable development is today viewed from the perspective of the three pillars model (Hopwood et al. 2005): economic, environmental and socio-cultural pillars respectively (Fig. 1). The economic dimension should build on the MDGs, which have helped to advance the world's agenda in the fight against poverty, hunger, and disease. The second pillar is environmental sustainability, usefully conceptualised by the global planetary boundaries. The third broad SDG is social inclusion, the commitment to future economic and technological progress under conditions of fairness and equitable access to public services, and with the government counter acting social discrimination on the basis of gender, ethnic origin, religion, and race.

Bettencourt and Kaur (2011) explain the importance of international and regional scientific organizations roles that need more of theoretical and methodological framing beyond the topics of individual research in the very area. Further development of sustainability science is described by Anand and Sen (2000), Kates (2011) and others. A country's economic, social and environmental performance is now seen through a framework of indicators of Sustainable Development, which were proclaimed in September 2015 during UN General Assembly. At the UN in New York the Open Working Group created by the UN General Assembly proposed a set of global SDGs which comprises 17 goals and 169 targets. Further to that, a preliminary set of 330 indicators was introduced in March 2015. Some SDGs build on preceding MDGs while others incorporate new ideas (Hák et al. 2016).² The successful implementation of Sustainable Development Agenda 2030 and its 17 goals requires scientific concept to be developed in order to monitor sustainable development in its three key areas: economic, environmental and social. When it comes to measuring sustainability it's often linked to well-being indicators, however they are not identical and shall be tackled differently (Neumayer 2010), Neumayer states, that fully integrated indicators of well-being and sustainability encounter a fundamental conceptual problem: those, that affect current well-being would not affect sustainability or not in the same way,

² The MDGs were derived from the United Nations Millennium Declaration, adopted by 189 nations in 2000. Most of the goals and targets were set to be achieved by 2015 on the basis of the global situation during the 1990s. The baseline for the assessment of progress is therefore 1990 for most of the MDGs targets.

and vice versa. Thus he proposes to complement Human Development Index (HDI) with appropriate sustainability indicators. The Sustainable Development Goals are the blueprint to achieve a better and more sustainable future for all. They address the global challenges we face, including those related to poverty, inequality, climate change, environmental degradation, peace and justice. The SDGs are interconnected, the potential synergies or trade-offs between the goals make the SDGs interactions complex, as they are clearly not independent from each other (Pradhan et al. 2017). However, the correlation does not imply causality, so often observed synergies between sustainable development indicators could be independently related to another process driving both indicators and in turn resulting in correlations. Therefore it becomes increasingly important to study those dependencies.

International interest in sustainability research has significantly increased in recent years due to boosts of the “Beyond GDP” initiative and the Stiglitz et al. (2017) report, and the idea of the SDGs has quickly gained ground because of the growing urgency of sustainable development for the entire world (Sachs 2012). Researchers stress the need to operationalise the SDGs’ targets and see an explicit need to structure the SDGs indicators into a coherent framework. They state that there is a need for global, integrated and scientifically based information on sustainability, and that the achievement of the SDGs need to be assessed and accompanied by target and indicators in a well structured framework. The path to quantifying and monitoring a multidimensional concept like sustainable development is fraught with challenges (Swain 2018). It requires a profound understanding of sustainable development, comprehension about how to operationalize and implement the SDGs, access to all forms of data and the expertise to analyse and interpret the results. Moreover, in his critical analysis of the SDGs, Swain writes that the ambitious UN-adopted SDGs have been criticized for being inconsistent, difficult to quantify, implement and monitor (Swain 2018). Disparaging analysis suggests that there exists a potential inconsistency in the SDGs, particularly between the socio-economic development and the environmental sustainability Goals (Spaiser et al. 2017; ICSU, ISSC 2015). Critiques also raise questions on if and whether the SDGs can be measured and monitored. Despite these criticisms, in these years, researchers have tried to measure and monitor sustainable development indicators, whose proposals are analyzed in detail in the Sect. 3. Starting from the successful implementation of Sustainable Development Agenda 2030 and from the existing Elementary Indicators (EIs) for the estimation of the single areas, we propose a methodological framework, estimated through to Higher-Order PLS-PM, of SDGs System, in which the hierarchical relationships between this different LVs are considered and their connections to a Higher-Order Global Sustainability CI is estimated.

This article does not seek to redefine the sustainability’s concept or propose new SDGs and/or targets. The aim of this paper is to propose PLS-PM as a method for studying existing SDGs indicators and demonstrate how it could help you to define the framework for SDGs indicators in order to provide a better measure of this complex multidimensional social phenomenon. There have been many and varied approaches in literature, such as simple arithmetic mean of the original indicators, PCA or FA to measuring sustainable development, based on simple aggregation techniques and easily calculated, but PLS-PM represents an important breakthrough with respect to these traditional aggregation methods. The choice of using the PLS-PM is useful for several reasons, particularly, it should be used for constructing a System of CIs, help to identify critical indicators, to construct ranking of countries and to provide an interpretation of the results. This approach allows you to estimate causal relationships, defined according to a theoretical model linking two or more latent complex concepts, each measured through a number of observable indicators (Lauro et al. 2018). The paper is organized as follow. Section 2 consists in a brief literature review

on different method proposed in the last few years for constructing sustainability indices. Section 3 propose a PLS-PM as a methodological framework for constructing a SDGs system. In this section advantages of this technique are discussed, in particular we explain how and why we can use PLS-PM and in detail the Higher-Order PLS-PM as a method for constructing SDGs composite indicators. Section 4 reports theoretical principles of PLS-PM approach and, in-depth, the Higher-Order PLS-PM and different approaches proposed in the literature. In Sect. 5 the constructing of the methodological framework for SDGs using Higher-Order PLS-PM is proposed, while Sect. 6 reports the main result of a case study for measuring SDGs of European Countries using PLS-PM approach. Finally, Sect. 7 conclusions are drawn and some open issues are explained.

2 Literature Review

The interest towards understanding and measuring the phenomena manifests in numerous researches and publications (Neumayer 2007; Stiglitz et al. 2017; Bettencourt and Kaur 2011; Sachs 2012; Gan et al. 2017; De Smedt et al. 2018). There have been many and varied approaches to measuring sustainable development. Many involve the identification of indicators of sustainable development and methods for collating these values, which are often indexed to allow for comparison (Evans et al. 2015). There is a significant body of international studies (Böhringer and Jochem 2007; Moran et al. 2008; Bilbao-Ubillos 2013; Estoque and Murayama 2014; Evans et al. 2015) aiming to review and compare the indices developed to measure sustainable development. However, there is still no consensus on a single index that is most acceptable by the scientific and political communities (Strezov et al. 2017). It is recognized, that measurement of sustainable development goals is somehow problematic due to its complicated nature (Spaiser et al. 2017; Fabbri et al. 2017). When it comes to measuring sustainability, it's often linked to well-being indicators or indicators that go beyond GDP measurement, however they are not identical and shall be tackled differently (Neumayer 2007) it states, that fully integrated indicators of well-being and sustainability encounter a fundamental conceptual problem: those, that affect current well-being would not affect sustainability or not in the same way, and vice versa. Thus he proposes to complement HDI with appropriate sustainability indicators. In general there are nine main indices, that measure sustainability to some extent, but only two of them Sustainable Society Index and Genuine Savings Index include all three areas: economic, social and environmental (Strezov et al. 2017). Those indices are easily calculated, based on simple aggregation techniques therefore can serve as excellent base for monitoring sustainable development on the cross country's level. However they do not demonstrate important interlinkages between goals and do show only observed variables. Current research shows big advances in reducing the dimensionality of the initial data by means of linear transformation techniques (Dunteman 1989; Smith 2002; Tenenhaus et al. 2005; Spaiser et al. 2017; Mazziotta and Pareto 2019) such as Principle Component Analysis (PCA). Those approaches help to understand social phenomena with large number of data, since they reduce dimensionality of the original indicator set. Nevertheless they are developed to examine the relationships between variables but not to determine weights. One of the difficulties here is that contradictory outcomes may be obtained, since the weights are correlation based and do not necessarily represent real tendencies among indicators (Hermans et al. 2008). Therefore Factor Analysis (FA) can attribute lower weights to important indicator due to their weak correlation with other dimensions. Spaiser et al. (2017)

use Confirmatory FA (CFA) to prove incompatibility of sustainable development goals on global scale. While it's important to show factors, that contribute to successful sustainable development but can not be directly observed or measured (health programmes, government investments, renewable energy and others) it's not clear how first and second order latent constructs are related in given study. There are other data-driven approaches, such as Bali Swain and Wallentin (2017) who, employing Structural Equation Models (SEM) to the Sachs et al. (2016) data set, compare their SDG construct with SDG indices from Sachs et al. (2016) and the HDI to investigate if these different measures suggest conflicting policy inferences to the developing and developed countries in terms of achieving SDGs (Swain 2018). In their analysis, the three pillars of SDGs are estimated in two steps and in the first step, the PCA is employed, while in the second step, Exploratory FA (EFA) is conducted to determine the SDGs that measure the latent factors or sustainable development pillars. According Swain (2018), the result obtained by (Bali Swain and Wallentin 2017) are in line with the literature that visualizes SDGs as an interlinked set of policies with trade-offs and synergies. As was stated before sustainability is often linked to well-being therefore we can as well study the use of PCA in order to select main factors (Mazziotta and Pareto 2019). The main problem with data-driven approaches is that, they lack theoretical foundation and consequently can send misleading signals to policy-makers, such an approach can be especially crucial for SDGs where multicollinearity is common therefore can be misleading.

3 An Alternative Proposal: Use of PLS-PM for Studying Sustainability

The goal of many research in social, economic and political fields is to obtain a whole description of the various facets of a complex phenomenon, through a suitable synthesis of the associated EIs. This combination can be obtained by applying methodologies known as *Composite Indicator (CI)* (Jarvis et al. 2003; Salzman 2003). The construction of a CI implies the search for a suitable synthesis of a number of Manifest Variables (MVs) in order to achieve a simple representation of a multidimensional phenomenon. Accordingly a CI can be considered as a latent concept, not directly measurable, the estimation of which can be obtained through the values of the MVs (i.e. the EIs). CIs are useful as a means of summarising the information provided by base indicators in an overall judgement or assessment of countries, and developing an assessment that permits countries to be ranked from best to worst, taking into account a whole set of features. Generally, they are very useful in order to deal with those phenomena that can not be observed directly. Different social concepts, such as the quality of life, satisfaction and sustainability, are multidimensional concepts that are difficult and complex to define. The main problem for researchers is to find appropriate tools to obtain a CIs able to synthesize and represent these phenomena. The existing literature offers different alternative approaches in order to obtain a CI (Trinchera et al. 2008). According Lauro et al. (2018) and Cataldo et al. (2017), PLS-PM should be used for constructing a System of CIs, help to identify critical indicators, to construct ranking of countries and to provide an interpretation of the results. PLS-PM allows you to estimate causal relationships, defined according to a theoretical model linking two or more latent complex concepts, each measured through a number of observable indicators. The basic idea is that the complexity inside a system can be studied by taking into account the entirety of the causal relationships among the LVs, each measured by several MVs (Lauro et al. 2018). PLS-PM represents an important breakthrough with respect to

traditional aggregation methods, such as a PCA or a simple arithmetic mean of the original indicators. Instead of taking the unweighted sum of the indicators (or unit-weights for all the indicators), PLS-PM assigns weights to the original variables taking into account the network of relationships between the constructs and the variance and covariance structure within and between the blocks of variables. Moreover, PLS-PM provides components with specific proprieties in order to enhance interpretation of the composites and the relationships among them. In particular, depending on the chosen estimation options, PLS-PM provides components that are as much correlated as possible to each other while explaining the variances of their own set of variables. PLS-PM is a procedure based on simple and multiple Ordinary Least Squares (OLS) regressions; it has minimal demands on measurement scales, sample size, and data distributions and it is particularly applicable for predictive applications and theory building (Chin 1998; Davino et al. 2018).

The choice of using the PLS-PM is particularly useful for several reasons. This approach has as its main advantages its applicability to small sample, the ability to estimate quite complex models (with many latent and observable variables) and less restrictive requirements concerning normality and variable and error distributions (Henseler et al. 2009). Furthermore, PLS-PM approach provides the possibility of working with missing data and in the presence of multi-collinearity. Another advantage of this approach, as compared to other multivariate techniques, is that it examines simultaneously a series of dependence relationship, using a single statistical approach to test the full scope of projected relations (Hair et al. 2009). Furthermore, this approach provides researchers with much more flexibility as it enables using both formative and reflective measurement models, providing a more nuanced testing of theoretical concepts (Hair et al. 2011). The PLS-PM approach should be used in order to not only reduce dimension but find relations between Composite Indicators and their blocks (Tenenhaus et al. 2005). However, a fundamental distinction must be made between reducing dimensionality and constructing composite indicators, because they are repeatedly confused. Both the procedures aims to summarize a set of variables or individual indicators, but reducing dimensionality focuses on extracting the most important information from the data, whereas constructing composite indicators focuses on the use of a measurement model that can be reflective or formative (Mazziotta and Pareto 2019). For a detailed explanation about composite indicators, formative and reflective measurement, see Mazziotta and Pareto (2019), Michalos (2014) and Nardo and Saisana (2008). For this purpose, it is essential to define the model of measurement in order to describe relationships between the phenomenon to be measured (LVs) and its elementary indicators (MVs).

According Mazziotta and Pareto (2019), measuring well-being requires a formative approach, where the index to be constructed does not exist as an independent entity, but it is a composite measure directly determined by a set of non-interchangeable individual indicators or pillars.³

Starting from these suggestions and observations and from the successful implementation of Sustainable Development Agenda 2030 and its 17 goals, we propose a System of SDGs Indicators estimated through the Higher-Order PLS-PM, in which the hierarchical relationships between this different LVs are considered.

³ One of the oldest and most famous formative composite indices is the HDI by United Nations Development Programme (United Nations Development Programme 2010). It is a composite measure of human development that include three theoretical dimensions (Health, Education and Income) (Mazziotta and Pareto 2019).

There are three main reasons for the inclusion of an Higher-Order Construct Model in PLS-PM. Firstly, by establishing Higher-Order Construct Models, researchers can reduce the number of relationships in the structural model, making the PLS path model more parsimonious and easier to grasp. Secondly, Higher-Order Construct Models prove valuable if the constructs are highly correlated; the estimations of the structural model relationships may be biased as a result of collinearity issues, and a discriminant validity may not be established. In situations characterized by collinearity among constructs, a Second-Order Construct can reduce such collinearity issues and may solve discriminant validity problems. Thirdly, establishing Higher-Order Construct Models can also prove valuable if formative indicators exhibit high levels of collinearity. Provided that the theory supports this step, researchers can split up the set of indicators and establish separate constructs in a higher order structure (Hair et al. 2014).

In the next section, the principles of PLS-PM approach is presented and the Higher-Order PLS-PM approach is described in detail.

4 Theory of PLS-PM

Two very important review papers on the PLS approach to SEM are Chin (1998) and Tenenhaus et al. (2005). Recently, Lauro et al. (2018) have presented some current developments in PLS-PM for the treatment of non-metric data, hierarchical data, longitudinal data and multi-block data.

4.1 The PLS Path Model

A PLS-PM is made up of two elements, the *measurement model* (also called the *outer model*), which describes the relationships between the MVs and their respective LVs, and the *structural model* (also called the *inner model*), which describes the relationships between the LVs. In the measurement model, an LV ξ is an unobservable variable (or construct) indirectly described by a block of observable variables x_k which are called MVs or indicators. There are three ways to relate the MVs to their LVs:

- The reflective way (or outwards directed way): each MV reflects the corresponding LV. In this case, the MVs should be highly correlated, due to fact that they are correlated with the LV of which they are expression. In other words, the block has to be homogeneous.
- The formative way (or inwards directed way): the LV is supposed to be generated by its own MVs.
Each MV or every set of MVs represents a different level of the underlying latent concept.
- The MIMIC way (a mixture of the reflective and formative ways): combination of the reflective and formative ways.

In the PLS-PM framework, the structural model specifies the relationships between the LVs; an LV, if it is supposed to depend on other LVs, is called exogenous, and, otherwise, endogenous. The evaluation of the measurement and structural model results in PLS-PM builds on a set of non-parametric evaluation criteria and uses procedures such as

bootstrapping and blindfolding. This process involves a separate assessment of the measurement model and the structural model.

A key characteristic of the PLS-PM method is the extraction of CI scores. One of the greatest advantages of PLS-PM is these CI scores. In the System of indicators built with PLS-PM, you can obtain the scores for each indicator, exogenous or endogenous, and for each indicator you can make a ranking among units. Moreover, PLS-PM provides information on the relative importance of constructs in explaining other constructs in the structural model. Information on the importance of constructs is relevant for drawing conclusions. For this reason, an Importance-Performance Matrix Analysis is a valuable decision making tool. The matrix contrasts the structural model's total effects (the importance) and the average values of the indicator (the performance). As a result, conclusions can be drawn on two dimensions (i.e., both importance and performance), which is particularly important in order to prioritize actions. The Importance-Performance Matrix Analysis is a very intuitive scatter plot and we can analyze themes according to the quadrant in which CIs are placed: upper-right quadrant: the *area to be maintained*, in which the CIs have a high value for the mean and the path coefficient; lower-right quadrant: the *area to improve* because the CIs have a high mean value and a low path coefficient; lower-left quadrant: the *area of the monitoring*, in which the CIs have a low value for the mean and the path coefficient; and upper-left quadrant: the *critical area*, because the CIs have a high impact but a low mean value. A similar scatter plot can be considered also for the MVs. In this kind of analysis, we have the possibility to analyze the strengths, weaknesses, opportunities, and threats of constructs, that are considered in the model in order to estimate a latent concept.

4.2 Higher-Order PLS-PM

PLS-PM is a suitable tool for the investigation of models with a high level of abstraction (Lohmöller 2013), in cases where the building of a system of Indicators depends on different levels of construction. Higher-order constructs in PLS-PM are considered as explicit representations of multidimensional constructs that exist at a higher level of abstraction and are related to other constructs at a similar level of abstraction, completely mediating their influence from or to their underlying dimensions (Chin 1998).

The Higher-Order Construct Models are also known as Hierarchical Models. As their name says, what makes these models so special is that they contain LVs of a *Higher Order*. A Higher-Order Construct is the fancy term that some authors use to refer to a special class of LVs: constructs that involve more than one dimension. Each of the Higher-Order Construct Models types is characterized by different relationships between the Higher-Order Constructs and the LVs: the reflective relationship and the formative relationship. There are four main types of Higher-Order Construct Models discussed in the extant literature (Wetzels et al. 2009) and used in applications (Johnson et al. 2012). These types of model have two elements: the *Higher-Order Construct*, which captures the more abstract entity, and the *Lower-Order Construct* which captures sub-dimensions of the abstract entity.

Different approaches have been developed and proposed in the literature: the Repeated Indicator Approach (Lohmöller 2013); the Two Step Approach (Sanchez 2013); the Mixed Two Step Approach (Cataldo et al. 2017); and the PLS Components Regression Approach (Cataldo et al. 2017).

The *Repeated Indicators Approach* is the most popular approach when estimating Higher-Order Constructs in a PLS-PM (Wilson 2009). The procedure consists of taking the

indicators of the Lower-Order Constructs and using them as the MVs of the Higher-Order LV.

The *Two Step Approach* is divided in two phases: in the first step the LV scores of the Lower-Order Constructs are computed without the Higher-Order Construct (Rajala and Westerlund 2010); then, in the second step, the PLS-PM analysis is performed using the computed scores as indicators of the Higher-Order Constructs. The implementation is not performed through a single PLS run; this implies that any Second-Order Construct, investigated in stage two, is not taken into account when estimating LV scores in stage one. Sanchez (2013) suggests this way of computing scores for the LVs of Lower-Order: we can obtain a score for a First-Order Construct by taking the first principal component of its indicators. Next, the PCA scores of the Lower-Order Constructs are subsequently used as indicators for the Higher-Order Construct in a separate PLS path model.

These two approaches present two important limitations related to components of each block: the first is that only one component is chosen for each block; the second is that this component has a strong representative power but a weak predictive power in the analysis of the Higher-Order Construct. For these reasons, in order to overcome these two drawbacks, two alternative methods to estimate the Second-Order Constructs are proposed (Cataldo et al. 2017). In particular, in order to resolve the issue related to the predictive power of the component for each First-Order Construct, the *Mixed Two Step Approach* is proposed and, regarding the choice of the number of components for each block, the *Partial Least Squares Component Regression Approach* is proposed.

The *Mixed Two Step Approach* begins with the implementation of the PLS-PM in the case of the Repeated Indicators Approach. So, initially, because the Second-Order Construct has no MVs of its own, we consider it as formed of all the MVs of the First-Order Constructs. Starting from this structure, a PLS-PM algorithm is performed in such a way as to obtain the scores of each block. Once the scores for the blocks have been obtained, these will be the MVs of the Second-Order Construct and the PLS-PM algorithm can be implemented. Compared to the Repeated Indicator Approach, this method uses the components that seems to be the best representative of its block and, at the same time, has the best predictive power on the Second-Order LV.

The *PLS Component Regression Approach* is proposed in order to overcome the problem related to the number of components of the First-Order Constructs, giving the possibility of choosing the number of components to be extracted manually or according to a criterion. In addition, since the aim of PLS-PM is to estimate the relationships between the LVs, this approach provides components that are at the same time representative of their blocks and predictive of the Second-Order Construct. In the first step of this approach, the PLS-Regression (Tenenhaus 1998) for each block of the First-Order is performed, so as to obtain h components for each block. Next, these h components will represent MVs of the Second-Order Construct. Once the PLS-Regression Components are assigned as indicators of the Second-Order Construct, the PLS-PM algorithm can be implemented. In this way, PLS-Regression for each block of the First-Order is performed, so as to obtain h components for each block. Next, these h components will represent MVs of the Second-Order Construct. For a detailed discussion and step-by-step illustration of these approaches, see Cataldo et al. (2017) and Hair et al. (2014). In this work the Higher-Order PLS-PM model has been estimated with the PLS Component Regression Approach (Cataldo et al. 2017). It has been demonstrated through a simulation study that this approach produce less biased and more stable parameter estimates than the approaches known in the literature (Cataldo et al. 2017). Moreover, the “plspm” package in the R statistical software (Sanchez and Trinchera 2012; Sanchez 2013) has been used.

5 A Methodological PLS-PM Framework for SDGs System

A sound methodological framework is the starting point in constructing CIs. The framework should clearly define the phenomenon to be measured and its sub-components, selecting individual indicators and weights that reflect their relative importance and the dimensions of the overall composite. This process should ideally be based on what is desirable to measure and not on which indicators are available (Joint Research Centre-European Commission 2008). In a framework, multi-dimensional concepts can be divided into several sub-groups. These sub-groups need not be (statistically) independent of each other, and existing linkages should be described theoretically or empirically to the greatest extent possible. According to PLS-PM approach, it is possible to define the SDGs as a multi-dimensional LVs not measurable directly and related to its single indicators or MVs by a formative relationship. The PLS-PM approach has enjoyed increasing popularity as a key multivariate analysis method in various research disciplines. It allows you to estimate causal relationships, defined according to a theoretical model linking two or more latent complex concepts, each measured through a number of observable indicators. The Global Sustainability model is reported in Fig. 2, that gives an overview of the approach used in the current research.

Global Sustainability is, here, conceived as a Third-Order Construct affecting Second-Order dimensions, which in turn shape First-Order LVs underlying specific aspects of the Second-Order dimensions. The study focuses on a formative-formative measurement model, a model resulting from the combination of formative lower order and formative

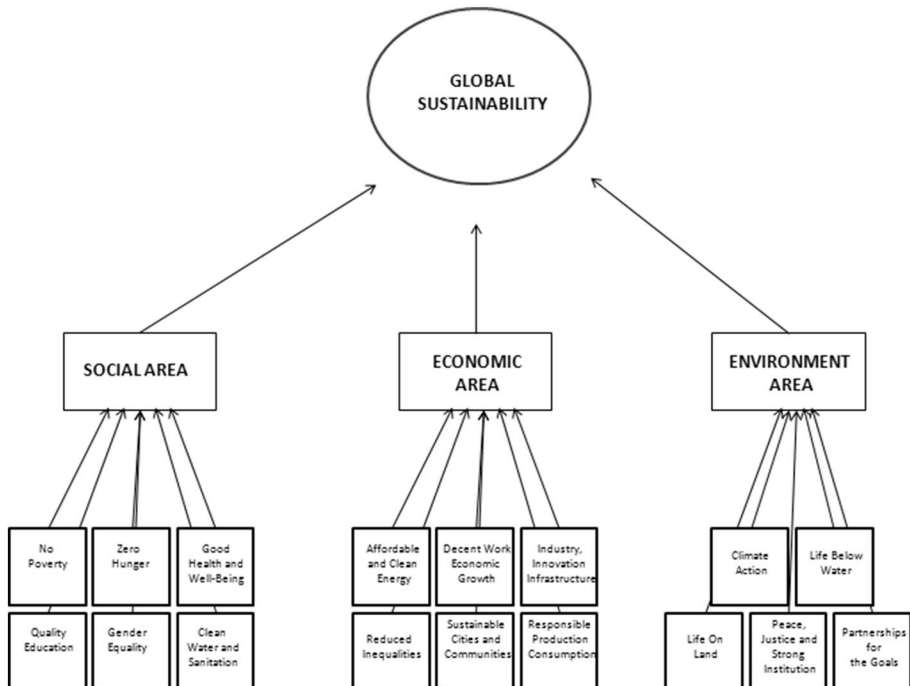


Fig. 2 The SDGs theoretical framework

Table 1 SDGs and the three areas

Area	Goals
<i>Social Area</i>	
Goal 1	No Poverty
Goal 2	Zero Hunger
Goal 3	Good Health and Well-Being
Goal 4	Quality Education
Goal 5	Gender Equality
Goal 6	Clean Water and Sanitation
<i>Economic Area</i>	
Goal 7	Affordable and Clean Energy
Goal 8	Decent Work and Economic Growth
Goal 9	Industry, Innovation and Infrastructure
Goal 10	Reduced Inequalities
Goal 11	Sustainable Cities and Communities
Goal 12	Responsible Production and Consumption
<i>Environment Area</i>	
Goal 13	Climate Action
Goal 14	Life Below Water
Goal 15	Life On Land
Goal 16	Peace, Justice and Strong Institution
Goal 17	Partnerships for the Goals

higher order constructs. This hierarchical structure allows us to synthesize individual indicators into single indexes in order to construct CIs at a global and a partial level. Moreover, we analyze the relationships among the different goals and areas as well as the effects of these on Global Index, in order to search for strongly influential factors. In Table 1, the Goals for each area are reported.⁴ The environmental dimension of sustainable development is measured with indices ranging between air quality, water quality, waste management, greenhouse gas emissions and various ecological indicators. The social dimension was the next most represented, with the widest variation in the selection of indicators. The economic dimension is generally measured by the capital income, including income distribution and public debt. There is no consensus or standard between the indices on the indicators used to measure each of the three dimensions of sustainable development (Strezov et al. 2017).

In the literature there is another SDGs classification made by the Overseas Development Institute's report (Nicolai et al. 2015). It classifies the SDGs into: reform, revolution and reversal. SDGs classified at reform levels are ending extreme poverty, strengthening economic growth in least-developed countries and halting deforestation (SDGs 1, 8, 15). Goals that require progress by multiples of current rates were categorized as revolution. Nine targets belong to this group: ending hunger, reducing maternal mortality, secondary school completion, ending child marriage, access to sanitation, access to energy (electricity), industrialization in less developed countries, reducing deaths and domestic resource

⁴ For a detailed description of the individual Goals, refer to the site "Sustainable Development Goals" (www.un.org/sustainabledevelopment/development-agenda/).

mobilization (SDGs 2–7, 9, 16, 17). Targets classified under reversals are moving in the opposite direction and require a reversal of current trends. They include inequality, slum populations, climate change, waste management and marine (reef) conservation. (SDGs 10–14). Nicolai and others are optimistic that the least-developed countries are halfway towards their 2030 targets of ending extreme poverty, economic growth in least-developed countries and halting deforestation (Swain 2018).

Clearly the ability of a CIs to represent multidimensional concepts largely depends on the quality and accuracy of its components. Missing data is a major problem in developing countries. For instance, even for well reported variables like child mortality, 136 of the 161 countries track data on this goal (Rodriguez-Takeuchi 2014). About two-thirds of the 75 countries accounting for more than 95% of all maternal, newborn and child deaths do not have records of births and deaths, whereas twenty-six countries have no data on child mortality since 2009 (Stuart et al. 2015). To facilitate the implementation of the global indicator framework, all indicators are classified by the IAEG-SDGs into three tiers based on their level of methodological development and the availability of data at the global level, as follows:⁵

- Tier 1: Indicator is conceptually clear, has an internationally established methodology and standards are available, and data are regularly produced by countries for at least 50 per cent of countries and of the population in every region where the indicator is relevant.
- Tier 2: Indicator is conceptually clear, has an internationally established methodology and standards are available, but data are not regularly produced by countries.
- Tier 3: No internationally established methodology or standards are yet available for the indicator, but methodology/standards are being (or will be) developed or tested.

Another important problem is concerning the high correlation of Goal's indicators to other Goals. Pradhan et al. (2017) highlight the existence of typically more synergies than trade-offs within and among the SDGs in most countries. This indicates a strong foundation for successfully implementation of the SDG agenda. The large fraction of synergies within each SDG is consistent with previously highlighted linkages between some indicators. An indicator may apply to more than one theme, as for example with “proportion of population with access to safe drinking water”, which has primary links to poverty and health. This indicator is also useful for measuring the impact of regulating and governing water utilities. Moreover, as domestic fresh water is the major source of drinking water for most countries, the indicator provides information on availability and use of water resources and for the availability of infrastructure in utilities (Pradhan et al. 2017). According some works, all SDGs need to act as a system of interacting cogwheels that together move the global system into the safe and just operating space. No SDG will do that individually, and the whole SDGs should not be seen as an additive structure but as a system of synergistic re-enforcement (Pradhan et al. 2017).

⁵ The latest changes reflect the decisions made during the IAEG-SDG WebEx Meeting in January 2019. The tier classification of many indicators is expected to change as methodologies are developed and data availability increases. The review of reclassification requests by the IAEG-SDGs will occur when requirements are met at the two physical meetings and via WebEx meetings throughout the year, based on a calendar developed by the Group.

6 A Case Study of SDGs of European Countries

To illustrate the importance of the proposal approach, in this section it is reported the analysis of SDGs framework. This section reports the main results of the Higher-Order PLS-PM analysis, with consideration on the evaluation of the measurement model, on the structural model and, finally, on the Importance-Performance Matrix Analysis. Also, the analysis of principal components has been carried out to compare this approach with the PLS-PM and to underline the advantages of the last one.

6.1 The Data and the Model

The official data derive from the database of United Nation “Sustainable Development Goals”.⁶ The analysis was developed with reference to the European Community countries, so the number of countries considered is 28. The data are updated to the last three years (2015–2017), in particular the data are related to the year 2017;⁷ but for some variables, because of the lack of data for that year, the previous years of this triennium were taken as the reference. Considering that each EIs have different units and values, for comparison purposes all units are first normalized to a value between 0 and 1, where 0 was assigned to the least sustainable while 1 was the value assigned to the most sustainable country for each EIs. The Eq. (1) is the simple expression used for normalization of each variable:

$$Z = \frac{X - \min(X)}{\max(X) - \min(X)} \quad (1)$$

where X is the sustainability index.

In the case of some EIs, where the larger value meant lower sustainability, the normalization had reverse order, so that the country with the highest EI received a value of 0, while the country with the lowest EI had a value of 1. The “Global indicator framework for the Sustainable Development Goals and targets of the 2030 Agenda for Sustainable Development” of UN 2017 (United Nations Statistical Commission 2017) lists all indicators used to calculate each of the selected SDGs. The indicators were assigned to one of the three dimensions of sustainable development: the economic, environmental and social dimensions.

In the analysis three Goals are not considered because they are not available in the three-year period considered. For the social area, the Goal not available is Goal 1 “No Poverty”; for the economic area, the Goals not available are “Reduced Inequalities” and “Responsible Production and Consumption”. The scheme of theoretical model that express the relationship among indicators is shown in Fig. 2 in Sect. 5. The study focuses on a formative-formative measurement model, a model resulting from the combination of formative lower order and formative higher order constructs. Sanchez “plspm” package in the R programming language (Sanchez and Trinchera 2012) was used in order to perform PLS-PM analysis involving formative indicators and the centroid scheme for the inner estimation.

⁶ <https://unstats.un.org/sdgs/indicators/database/>.

⁷ Data presented in this study were mined in September 2018.

Table 2 Correlation among Goals of Social Area

	Zero Hunger	Good Health	Quality Education	Gender Equality	Clean Water	Social Area
Zero Hunger	1.00					
Good Health	0.82	1.00				
Quality Education	0.89	0.86	1.00			
Gender Equality	0.55	0.52	0.69	1.00		
Clean Water	0.87	0.79	0.97	0.67	1.00	
Social Area	0.92	0.89	0.98	0.74	0.96	1.00

Table 3 Correlation among Goals of Economic Area

	Clean Energy	Economic Growth	Infrastructure	Sustainable Cities	Economic Area
Clean Energy	1.00				
Economic Growth	0.90	1.00			
Infrastructure	0.54	0.56	1.00		
Sustainable Cities	0.74	0.67	0.46	1.00	
Economic Area	0.96	0.94	0.46	0.79	1.00

Table 4 Correlation among Goals of Environmental Area

	Climate Action	Life Below Water	Life On Land	Environmental Area
Climate Action	1.00			
Life Below Water	0.99	1.00		
Life On Land	0.99	0.99	1.00	
Environmental Area	0.99	0.99	0.99	1.00

6.2 The Main Results

Tables 2, 3 and 4 show the correlation matrix of Sustainable Goals for each area and between each of them and their own area.

As we can see from the tables, for each area, the Goals are strongly related to each other and strongly linked to the area indicator. Only “Gender Equality” for Social area and “Infrastructure” for Economic area are weakly correlated with the other goals of the same area. In general, this means that the Goals of different dimensions of sustainability are, with different intensity, concordant among themselves. Before applying the PLS-PM approach, an exploratory PCA has been performed in order to analyse the relationship of elementary indicators for each Goal. In Fig. 3 the scree-plot (a) and the correlation circle (b) of PCA of Goal 1 “No Poverty” are shown.

From the scree-plot examination, it is evident that the first one component explain the variability of the MVs but it is not sufficient to synthesize the block of MVs of the

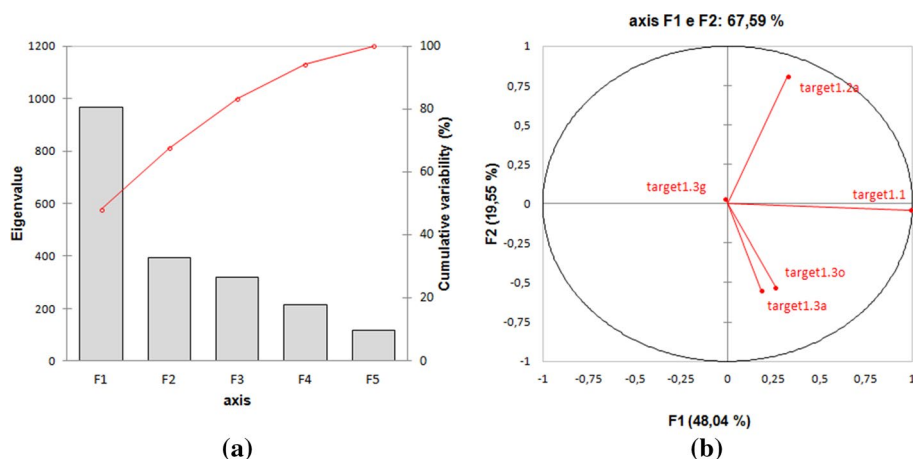


Fig. 3 Scree-plot and correlation circle of PCA of Goal 1 “No Poverty”

Table 5 Godness of Fit

GoF	Relative GoF	GoF outer model	GoF inner model
0.656	0.762	0.718	0.769

Table 6 Overall model quality

	Type	R^2	Communalities	Redundancies
Social Area	Endogenous	0.934	0.818	
Economic Area	Endogenous	0.916	0.600	
Environment Area	Endogenous	0.895	0.998	
Global Sustainability	Endogenous	0.991		0.636

Goal. The most of the variability of the MVs of Goal 1 (67.59%) can be explained by the first two axis. By projecting the original indicators in the plane of the first two principal components, the circle of correlations is obtained, where each indicator of Goal is represented by a point with coordinates equal to the two coefficients of correlation with the first and second factor. Note that the first factor is strongly correlated with one indicator of 5, whereas the second one represents the composite index of the other three indicators. The graph shows that an indicator is in the center of the circle, this means that it is not explained by the first two factors. The same analysis has been carried out for all 17 Goals and, more or less, the same results came out. In this way it is possible to have several main components representing the different goals but they can be analyzed separately from each other. This means that for this type of sustainability data the PCA approach can be used for single each block, but it is unable to summarize the overall structure of the data. For this reason, since the goal of this work is not only to synthesize the dataset, but to obtain a composite indicator capable of representing this phenomenon, the PLS-PM approach has been applied and the results are presented below. First, overall model quality indices were

Table 7 Path coefficients linking the first-order constructs to the three areas at second-order construct and path coefficients linking the three areas to Global Sustainability CI

LVs	Path coefficient	SE	CI (95%)
Zero Hunger → Social Area	0.417	0.025	[0.394; 0.465]
Good Health → Social Area	0.408	0.033	[0.332; 0.429]
Quality Education → Social Area	0.441	0.067	[0.277; 0.465]
Gender Equality → Social Area	0.375	0.018	[0.358; 0.409]
Clean Water and Sanitation → Social Area	0.458	0.062	[0.424; 0.620]
Clean Energy → Economic Area	0.505	0.071	[0.390; 0.590]
Economic Growth → Economic Area	0.569	0.086	[0.455; 0.704]
Infrastructure → Economic Area	0.523	0.032	[0.368; 0.651]
Sustainable Cities → Economic Area	0.465	0.034	[0.392; 0.499]
Climate Action → Environmental Area	0.537	0.080	[0.532; 0.741]
Life Below Water → Environmental Area	0.529	0.023	[0.492; 0.567]
Life On Land → Environmental Area	0.536	0.013	[0.531; 0.569]
Social Area → Global Sustainability	0.312	0.0939	[0.288; 0.683]
Economic Area → Global Sustainability	0.438	0.0658	[0.356; 0.760]
Environmental Area → Global Sustainability	0.241	0.0120	[0.162; 0.398]

shown. The coefficients presented in Table 5 showed that MVs and LVs are well predicted by the PLS-PM framework. The prediction performance of the PLS-PM is high for both the outer, inner and global models. As concerning the goodness of fit, there is no overall fit index in PLS-PM. Nevertheless, a global criterion of Goodness of Fit (GoF) has been proposed by Tenenhaus et al. (2005). Both GoF and relative GoF reflected the high quality of the constructs for both outer and inner model (Table 5).

Table 6 reports the main indices indicating the overall model quality: R^2 coefficient, the Communality and Redundancy indices. The R^2 coefficient shows that endogenous LVs are well predicted by the explanatory LVs, while the values of communality index (another index that measures the goodness of the models of measurement) are appreciably higher for all blocks (value of 0.50 indicates a sufficient degree of construct validity).

In order to provide an interpretation of the results, we analyse all the path coefficients linking the first-order constructs to the three areas at second-order construct and path coefficients linking these areas to Global Sustainability CI through Table 7.

The path coefficients are all positive and significant. All dimensions of a first-order construct are important for their areas, respectively. In particular the dimensions of “Quality Education” and “Clean Water and Sanitation” prove to be most influential among all the factors of the social area; the most influential dimension of economic area is “Economic Growth”, while the “Climate Action” has a slightly higher path coefficient than the other dimensions of the Environmental Area. The economic area has a significant role in the Global Sustainability CI estimation, with a path of 0.438, following by the social area, with a path of 0.312. The environmental area has a lower impact to estimation Global Sustainability CI, with a path of 0.241.

The importance played by Goals on the outcome domain can be jointly analysed with the composites averages. Figure 4 shows the importance and the performance (means and path-coefficients) of the 12 Goals in order to estimate the different areas of Global Sustainability CI.

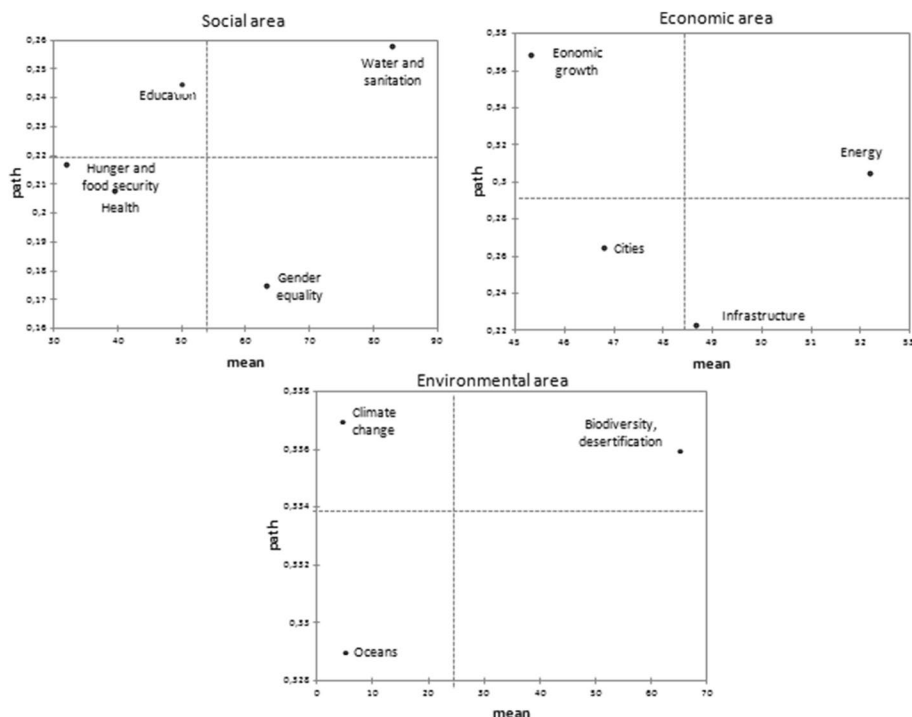


Fig. 4 Importance-performance matrix analysis

The Importance-Performance Matrix Analysis shows that the Education, Economic growth and Climate change are respectively located in a critical area of the three areas respectively, because the critical area contains domains that have a high impact on the outcome but that register low average values; they prove to be especially critical for the Global Sustainability CI estimation. Instead “Zero Hunger”, “Good Health”, “Sustainable Cities” and “Life Below Water” are located in a area of the monitoring. Note how the “Gender Equality” and “Infrastructure” are placed in the area to improve.

This application can allow only a provisional and preliminary study. Because of the missing data, some Goals were not considered in the analysis so it is incomplete and partial. The results presented only want to be example to show the proposed instrument and highlight its potential. According the United Nations site, starting 2019 major updates are expected to be released in July, September and December, so maybe it will be possible have Goals that in this analysis were not considered due to the missing EIs. Moreover, in this analysis we consider all indicators used on 28 European countries on the same level, but we should avoid the sustainability indices/indicators tend to over- or under-estimate developed countries as compared with developing countries.

7 Final Remarks

The development of a quality framework for composite indicators is not an easy task. In fact, the overall quality of the composite indicator depends on several aspects, related both to the quality of elementary data used to build the indicator and the soundness of the procedures used in its construction. Quality is usually defined as “fitness for use” in terms of user needs. The most important quality characteristics depend on user perspectives, needs and priorities, which vary across user-groups (Joint Research Centre-European Commission 2008).

Lack of limited availability of the data is a major constraint for quantifying and monitoring SDGs. While data on economic indicators is widely available for most countries, data on environmental and social indicators is incomplete and of poor quality. Furthermore SDGs have often been criticized for a lack of underlying comprehensive theoretical framework. Finally, SDGs are long-term development agenda and have the potential to be exposed to unforeseen positive and negative shocks. There have been many and varied approaches, such as simple arithmetic mean of the original indicators, PCA or FA to measuring sustainable development, based on simple aggregation techniques and easily calculated. According to many authors, such as Mazziotta and Pareto (2019), the construction of index for measuring this multidimensional phenomenon cannot solve simply by using PCA or FA, since they are typically used for a reflective approach and in particular a PCA based index accounts for a limited part of the total variance. PLS-PM represents an important breakthrough with respect to these traditional aggregation methods. The choice of using the PLS-PM is useful for several reasons, particularly this approach should be used in order to not only reduce dimension but find relations between composite indicators and their blocks (Tenenhaus et al. 2005). It should be used for constructing a system of composite indicators, help to identify critical indicators, to construct ranking of countries and to provide an interpretation of the results. PLS-PM allows you to estimate causal relationships, defined according to a theoretical model linking two or more latent complex concepts, each measured through a number of observable indicators. This approach, as compared to other multivariate techniques, examines simultaneous a series of dependence relationships, using a single statistical approach to test the full scope of projected relations (Hair et al. 2009). The basic idea is that the complexity inside a system can be studied by taking into account the entirety of the causal relationships among the LVs, each measured by several MVs (Lauro et al. 2018). It is absolutely essential to define the theoretical framework with an appropriate measurement model. This paradigm should always be considered when the objective of the research is to measure a multidimensional phenomenon through composite indices. The PLS-PM provides researchers with much more flexibility as it enables using both formative and reflective measurement models, providing a more nuanced testing of theoretical concepts (Hair et al. 2011). This present work would be a first propose of a methodological framework that can be useful for creating a Sustainability CI in terms of decision-making. As a first approach, we analyse the PLS-PM approach without considering synergies and trade-offs within an SDGs. We propose Higher-Order PLS-PM approach for studying existing SDGs indicators and demonstrate how it could help you to define the methodological framework for SDGs system in order to provide a better measure of this complex multidimensional social phenomenon. A Higher-Order PLS-PM approach was adopted in a preliminary attempt to investigate relations between different analytical dimensions of the phenomenon. The results of the analysis suggest that economic area should be the most important in order to measure sustainability but it cannot be considered the only factor to

be analyzed, as the other two dimensions also have their relative impact. However, our preliminary research can offer only suggestions on this aspect and further investigation are definitively needed, especially because some Goals were not considered in the analysis due to missing data. Starting from this methodological PS-PM framework for SDGs system and works of Pradhan et al. (2017), future works will be to implement and develop these models also considering synergies within SDGs and understand if and, if yes, how these synergies modify the relationships among the Goals and the areas. Moreover, we would like to insert in the analysis qualitative information concerning European community countries in order to understand if there are differences among them relating to this issue. The successful implementation of the good methodological framework for SDGs system is the only way forward to address the global sustainability challenge of ensuring human well-being, economic prosperity, and environmental protection. Infact, “what we measure affects what we do; and if our measurements are flawed, decisions may be distorted” (Stiglitz et al. 2017).

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